

GIANT: Structure-Agnostic Practical Adversarial Attacks for Graph-based Network Intrusion Detection Systems

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Abstract

Graph-based Network Intrusion Detection Systems (GNIDS) have emerged as a promising solution by capturing sophisticated interaction patterns through inter-flow analysis. However, their robustness remains a critical yet unexplored issue. Despite efforts devoted to GNIDS robustness evaluation, most of them exclusively focus on feature attacks while neglecting topological vulnerabilities. Moreover, many studies either oversimplify the adversary by adopting random topology perturbations, or conversely assume unrealistic adversary's knowledge and capabilities, such as privileged network access or complete graph awareness. The lack of practical robustness evaluation severely hinders the deployment of GNIDS in real-world security applications.

To fill this gap, we propose GIANT, a structure-agnostic practical adversarial attack framework for comprehensive robustness evaluation of GNIDS. Different from prior methods that perturb flow features or presuppose a fixed graph construction mode, GIANT

injects extra adversarial network flows without altering existing traffic data to jointly manipulate graph topology and flow features, ensuring transferability across diverse GNIDS. Specifically, GIANT first transforms network flows into a hypothetical line graph and then performs a two-phase attack to determine adversarial flow endpoints and optimize adversarial flow features, balancing maximum adversarial impact and stealthiness. The iterative injection of adversarial flows induces erroneous decisions in the target GNIDS. Extensive experiments on two public datasets covering IoT and cloud environments validate GIANT's effectiveness, transferability, and efficiency against existing attack methods, providing a practical robustness evaluation solution for GNIDS, and offering critical insights into their fundamental vulnerabilities.

CCS Concepts

• Security and privacy → Intrusion detection systems.

Keywords

Network Intrusion Detection Systems, Adversarial Attacks, Graph Neural Networks

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1 Introduction

Network Intrusion Detection Systems (NIDS) [5, 11, 39] play a pivotal role in enterprise security infrastructures, constantly monitoring network communications to uncover suspicious patterns and generating alerts for security analysts. Traditional NIDS (e.g., Snort and Suricata) typically operate at the flow level, inspecting packet features and statistical patterns to match rule-defined anomalies. However, as threat actors increasingly coordinate multiple flows across different hosts to perform stealthy threat activities, intra-flow analysis methods [1] gradually become invalid. Recently, Graph-based NIDS (GNIDS) have emerged as a promising solution by modeling sophisticated interactions among network entities as graphs at various levels (e.g., flow, endpoint, host), enabling multi-granularity inter-flow analysis [14, 24, 42] to capture the topological patterns indicative of potential threats.

Motivation. Despite the remarkable progress of GNIDS, a critical gap remains: their robustness under challenging conditions is still largely unexplored. Prior works on GNIDS [17, 32] have mainly focused on improving detection accuracy under benign settings, while paying limited attention to their behavior when facing *adaptive* or *intentionally* manipulated traffic. Empirical studies on GNNs [15, 31] have shown that even minor manipulations, such as inserting a single node or edge, can mislead the model's decision boundaries. Such vulnerability raises serious concerns about whether existing GNIDS can remain reliable when confronted with increasingly sophisticated evasion tactics. Therefore, systematically investigating the resilience of GNIDS against realistic manipulations is not only necessary but also crucial for ensuring their trustworthy deployment in real-world network environments.

Existing Limitations. While numerous adversarial attacks have been developed for general GNNs [15, 31], directly applying them to GNIDS is non-trivial due to unique operational constraints and threat models (see Section 2 for details).

Although a few recent studies [34, 38, 42] begin to study the robustness of GNIDS, current efforts still exhibit three key limitations that hinder their practical relevance: (1) **Lack of joint consideration for feature and topology manipulations:** Existing GNIDS methods [6, 7] primarily emphasize perturbations on traffic features (e.g., packet lengths or inter-arrival times) while overlooking vulnerabilities in graph structures. This one-sided focus leaves GNIDS unprotected against attacks exploiting the relational dependencies among flows, leading to incomplete robustness evaluation. (2) **Over-simplistic perturbation strategies:** Although there are a few GNIDS studies that consider topology-oriented attacks [34, 35], the majority adopt random or brute-force perturbation schemes—such as arbitrarily adding or removing nodes, edges, or features—under fixed graph construction settings. These coarse-grained strategies fail to reflect the real level of adversaries and result in inconsistent or unrepresentative robustness evaluation of targeted GNIDS. (3) **Unrealistic and overstated threat models:** Existing attacks often rely on idealized assumptions that are rarely achievable in real network environments. For feature-level attacks [10, 20, 27], the adversary is typically assumed to intercept and directly alter

existing traffic, requiring privileged network access that genuine attackers seldom possess. For topology-level attacks [41], it is further assumed that the adversary has full visibility of the entire communication graph. Such unrealistic premises undermine the credibility and practical value of existing robustness evaluations.

Our Work. Given the above limitations, we summarize six core requirements to develop a practical adversarial attack to evaluate the robustness of GNIDS (details are in Appendix A), including *effectiveness* (produce measurable degradation of detection outcomes), *efficiency* (achieve strong impact with a small injection budget and modest runtime), *practicality* (operate without altering existing traffic or assuming privileged access), *stealthiness* (keep injected flows consistent with benign patterns to avoid trivial filtering), *scalability* (apply to large, dynamic traffic graphs with low overhead), and *transferability* (generalize across heterogeneous GNIDS).

In this study, we propose **GIANT**, a practical attack for comprehensive GNIDS robustness evaluation. Unlike prior attacks [26, 28, 30, 35] that perturb existing flow features or rely on a fixed graph construction method, GIANT is a *structure-agnostic* attack that neither presupposes any particular graph structure or construction method. It injects additional adversarial flows on a reversible flow-centric representation to jointly manipulate topology and flow attributes, enabling unified evaluation across diverse GNIDS. Guided by the six design requirements, GIANT achieves *effectiveness* by injecting flows that strategically disrupt decision boundaries, ensures *efficiency* through a budget-aware and iterative design, preserves *practicality* via an injection-only threat model consistent with real adversaries, maintains *stealthiness* by constraining injected features within normal traffic patterns, *scales* to large graphs through a lightweight two-phase process, and attains *transferability* by operating on a reversible line-graph view applicable to heterogeneous GNIDS.

In summary, the contributions of our paper are threefold:

- We propose GIANT, the first structure-agnostic adversarial attack framework for robustness evaluation of GNIDS. By operating on a hypothetical line graph, GIANT avoids information misalignment across distinct graph construction modes, enabling structure-agnostic and transferable attacks across diverse GNIDS.
- We develop a two-phase attack strategy to inject adversarial network flows by endpoint determination with heuristic rules and feature generation with gradient guidance from a surrogate model. This flexible attack strategy supports both on-path and off-path adversary assumptions to meet realistic conditions.
- We conduct extensive experiments on two widely used datasets spanning IoT and cloud environments to demonstrate the effectiveness, transferability, and efficiency of GIANT against existing state-of-the-art GNIDS, significantly outperforming other adversarial attack methods.

2 Preliminaries

2.1 Threat Model

In this paper, we investigate practical and general adversarial attacks against graph-based network intrusion detection systems (GNIDS). The threat model is defined along three dimensions: the adversary's goal, knowledge, and capability.

Adversary's Goal: The primary goal of the adversary is to evade detection at test time. Both poisoning attacks during the training

phase and trivial evasion attempts against intra-flow analysis are beyond the scope of this work. Specifically, the adversary aims to strategically modify the graph topology by injecting adversarial flows through compromised hosts, thereby misleading GNIDS into misclassification.

Adversary’s Knowledge: We assume a black-box setting follows common assumptions in [35], where the adversary has access to the training data but lacks knowledge about the internal architecture, parameters, or specific graph construction mode of the target GNIDS. During the testing phase, the adversary’s knowledge can manifest in two settings: *on-path* where the adversary is able to monitor complete network traffic and hence reconstruct the full graph topology, and *off-path*, where visibility is limited to the sub-graph involving compromised hosts. A more detailed discussion of the distinction will be provided in Section 3.3.

Adversary’s Capability: To ensure the practicality, the adversary’s capability is constrained to align with practical conditions. Specifically, the adversary is not allowed to modify existing traffic data. Instead, the adversary can only generate additional adversarial flows from compromised hosts to alter the local or global graph topology, leveraging the topological vulnerabilities to influence GNIDS decisions.

2.2 Problem Formulation

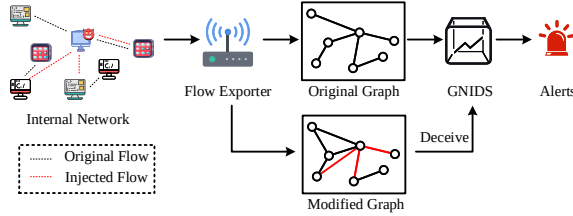


Figure 1: The overall deployment architecture of GIANT.

As stated in the previous subsection, GIANT seeks to inject adversarial flows through compromised hosts rather than modify existing traffic data, thereby altering the graph topology to influence GNIDS decisions. The attack process is to first train a surrogate model to approximate the target GNIDS, then inject adversarial flows to attack the surrogate model, and finally transfer them to the target model. The overall deployment architecture of GIANT is illustrated in Figure 1.

Specifically, given a target GNIDS, we construct a line graph $\mathcal{G}_{\text{train}}$ from the training dataset, where nodes denote network flows, and edges connect flows with common endpoints. A surrogate model $\mathcal{M}$ is trained for node classification to obtain gradient information, formulated as:

$$\theta^* = \arg \min_{\theta} \mathcal{L}_{\text{train}}(f_{\theta}(\mathcal{G}_{\text{train}})), \quad (1)$$

where f_{θ} denotes the surrogate model parameterized by θ , and $\mathcal{L}_{\text{train}}$ denotes the training loss.

During inference, for each batch of traffic, we construct a line graph $\mathcal{G} = (V, A, X)$, where V denotes the node set, A denotes the adjacency matrix, and X denotes the node features. Then we inject adversarial flows to modify $\mathcal{G}$ into $\mathcal{G}' = (V', A', X')$ with

$$A' = \begin{bmatrix} A & V_I \\ V_I^T & A_I \end{bmatrix}, \quad X' = \begin{bmatrix} X \\ X_I \end{bmatrix}. \quad (2)$$

where A_I denotes the adjacency matrix of the injected nodes, V_I denotes a matrix that records the connections between the original and the injected nodes, and X_I denotes the feature matrix of the injected nodes.

The adversarial objective is to maximize the misclassification probability of the modified graph $\mathcal{G}'$ while preserving attack stealthiness, formulated as:

$$\mathcal{G}'^* = \arg \min_{\mathcal{G}'} \mathcal{L}_{\text{atk}}(f_{\theta^*}(\mathcal{G}')), \text{ s.t. } \|\mathcal{G}' - \mathcal{G}\| \leq \epsilon, \quad (3)$$

where $\mathcal{L}_{\text{atk}}$ is typically defined as $-\mathcal{L}_{\text{train}}$, and ϵ controls the perturbation budget.

Finally, we transfer $\mathcal{G}'^*$ to other target GNIDS to realize adversarial attacks.

3 Methodology

3.1 Overview of GIANT

In this section, we propose GIANT, a novel and practical Graph Injection Attack for Network Traffic to comprehensively evaluate the robustness of Graph-based Network Intrusion Detection Systems (GNIDS).

The workflow of GIANT is shown in Fig. 2, which consists of three key steps. We briefly introduce the design and motivation of them. In (1) graph transformation, we periodically retrieve network flows and represent them as a graph to model the complex interactions between hosts. GNIDS may employ diverse graph construction modes (e.g., hosts, endpoints, or flows as nodes). To remove ambiguity from parallel edges and unify different formats, we linearize the original graph into a hypothetical line graph. During this process, flows are translated into nodes, connected if they share a common endpoint. The transformation turns an edge classification task on multigraphs into a node classification task on simple graphs, simplifying the subsequent attack process. Notably, we maintain a table that records the endpoint pair of each flow to ensure reversible transformation. To realize practical attacks inline with realistic constraints, we perform graph injection attacks on line graphs to generate adversarial network flows, which comprises of (2) endpoint determination and (3) feature optimization. As the line graph topology is naturally determined, we select the most susceptible flows to first determine the endpoints of flows to be injected by confidence score analysis and node degree analysis. Then, to maximize the damage of adversarial flows with stealthy constraints, we refine the flow features with a devised loss function. So far, we have generated adversarial network flows sequentially for robustness evaluation. The reversible transformation ensures that our attack can transfer and generalize to various GNIDS with different graph construction modes.

3.2 Graph Transformation

GNIDS periodically collect traffic data from internal networks and split them into 5-tuple flows for subsequent analysis, commonly leveraging NetFlow/IPFIX logs [12] as the data source. Existing GNIDS solutions have developed various graph construction modes

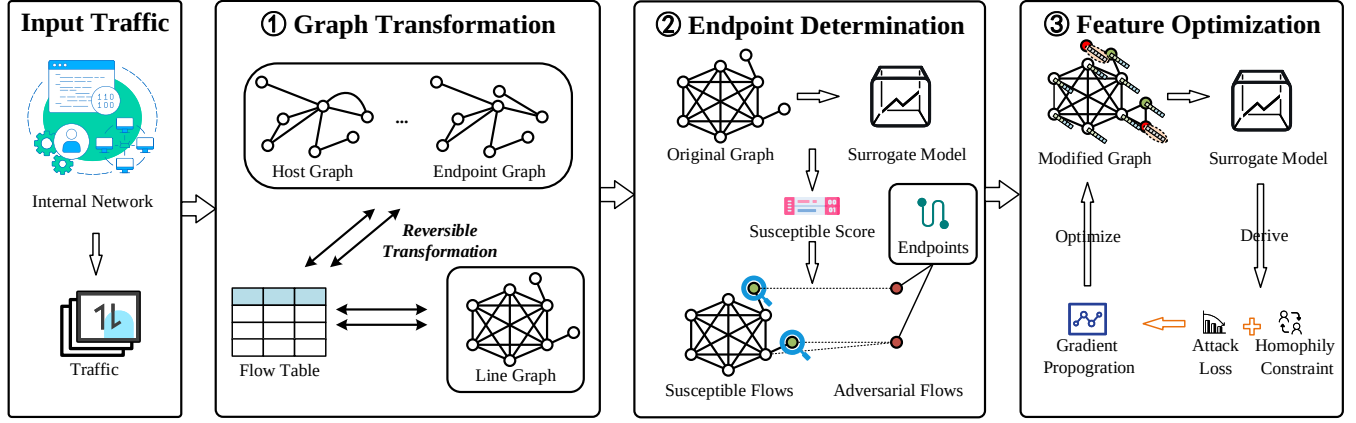


Figure 2: The overall framework of GIANT.

for multi-granularity inter-flow analysis. The most commonly used schemes include:

- **Endpoint Graph:** Nodes denote endpoints (i.e., IP-port pairs), and edges denote flows between pairs of endpoints.
- **Host Graph:** Node denote hosts (i.e., IP addresses), and edges denote flows between pairs of hosts.
- **Line Graph:** Node denote flows, and edges indicate at least one shared endpoint between pair of flows.

Notably, GNIDS built on endpoint/host graphs frame intrusion detection as an edge classification task, whereas those on line graphs formulate it as a node classification task. A critical challenge concerning endpoint/host graphs is that there may exist multiple communications between endpoints/hosts, which naturally results in parallel edges and thus forms multigraph structures. This structural feature introduces substantial complexity for designing adversarial attacks, as perturbations (e.g., adding/modifying edges) must account for redundant topological information, complicating the decision-making process for adversarial attacks.

To eliminate ambiguity introduced by parallel edges and to unify heterogeneous graph construction modes, we linearize the constructed multigraph into a hypothetical line graph. We maintain a flow table that records the endpoint pairs and associated features, enabling reversible transformation between different graph formats.

Additionally, the line graph format provides advantages for both adversarial attacks and defenses in GNIDS. From the attack perspective, it transforms parallel edges into distinct nodes, simplifying complex edge perturbations (e.g., adding redundant flows between a pair of endpoints) into straightforward node injection operation, directly corresponding to flow injection operation, which is practical to make decisions and easily implementable.

From the defense perspective, the line graph allows free node/edge deletion without violating practical constraints. In most GNIDS solutions, the final flow representation are obtained by concatenating endpoint embeddings. This design allows the defender to remove adversarial flows and tailor the line graph by deleting nodes and edges, while preserving consistency across different graph construction modes, provided that the endpoint nodes remain intact.

Thus, the line-graph formulation provides a principled and practical basis for systematic robustness studies of GNIDS.

3.3 Endpoint Determination

As the line graph topology is inherently derived from the source and destination addresses of network flows, to maximize the adversarial impact of injected flows against GNIDS, we first determine their both endpoints to establish connections with the most vulnerable flows. Typically, the aggregation process of a GNN layer for node v can be denoted as:

$$\mathbf{h}_v^k = \phi \left(\mathbf{h}_v^{k-1}, \mathbf{f} \left(\left\{ \mathbf{h}_u^{k-1} \right\}_{u \in N(v)} \right) \right), \quad (4)$$

where $\mathbf{f}(\cdot)$ denotes a neighborhood aggregation function, $\phi(\cdot)$ denotes a feature update function, $N(v)$ denotes the neighborhood of node v , and $\mathbf{h}_v^k$ denotes the hidden representation of node v at layer k .

GNNs stack multiple layers to capture multi-hop neighborhood information, where only one-hop neighbors are involved in each layer. However, from the attackers' perspective, the aggregation process shall be decomposed into layer-wise contributions to explicitly quantify the adversarial impact of an injected node. In this way, $N(v)$ can be extended to include both one-hop and multi-hop neighbors of node v . Let $N_t(v)$ and $\mathbf{f}_t(\cdot)$ denote the t -hop neighborhood of node v , and corresponding aggregation function. Eq. (4) can be then expanded as:

$$\mathbf{h}_v^k = \phi \left(\mathbf{h}_v^{k-1}, \mathbf{f}_1 \left(\left\{ \mathbf{h}_u^{k-1} \right\}_{u \in N_1(v)} \right), \dots, \mathbf{f}_n \left(\left\{ \mathbf{h}_u^{k-1} \right\}_{u \in N_n(v)} \right) \right). \quad (5)$$

With the injection of adversarial flows, N_t changes to N'_t , and $\mathbf{f}_t$ changes to $\mathbf{f}'_t$, i.e., $\mathbf{f}'_t - \mathbf{f}_t = \Delta \mathbf{f}_t$. We apply a Taylor expansion of the multivariable functions to approximate the updated representation $\mathbf{h}_v^k$ for node v at layer k as:

$$\begin{aligned} \mathbf{h}_v^k = & \mathbf{h}_v^k + \frac{\partial \phi}{\partial \mathbf{f}_1} (\mathbf{f}'_1 - \mathbf{f}_1) + \dots + \frac{\partial \phi}{\partial \mathbf{f}_n} (\mathbf{f}'_n - \mathbf{f}_n) \\ & + O((\Delta \mathbf{f}_1)^2 + \dots + (\Delta \mathbf{f}_n)^2), \end{aligned} \quad (6)$$

where the second-order term reflects higher-order perturbation effects. By denoting $\frac{\partial \phi}{\partial \mathbf{f}_t}$ at layer k as $\mathbf{p}_{t,k}$, we have a simplified expression as:

$$\Delta \mathbf{h}_v^k = \mathbf{h}_v^{k*} - \mathbf{h}_v^k = \sum_{t=0}^n \mathbf{p}_{t,k} \Delta \mathbf{f}_t + o\left(\sum_t \Delta \mathbf{f}_t\right), \quad (7)$$

where $o(\cdot)$ denotes asymptotically negligible terms under small perturbations. Without loss of generality, we assume $\mathbf{f}_t(\cdot)$ to be a widely-used weighted sum aggregator as:

$$\mathbf{f}_t\left(\{\mathbf{h}_u^{k-1}\}_{u \in N_t(v)}\right) = \sum_{u \in N_t(v)} w_{u,t} \mathbf{h}_u^{k-1}, \quad (8)$$

where $w_{u,t}$ encodes the topological weight within t -hop neighborhood of node v . Therefore, we can conduct t -hop node injection to change $N_t(v)$ to $N'_t(v)$ and to perturb $\mathbf{f}_t$ as:

$$\Delta \mathbf{f}_t = \sum_{u \in N'_t(v)} \left(\Delta w_{u,t} \mathbf{h}_u^{k-1} + w_{u,t} \Delta \mathbf{h}_u^{k-1} \right). \quad (9)$$

The impacts of original and injected neighbors differ. For original neighbors $u \in N_t(v)$, their features $\mathbf{h}_u$ preserve, i.e., $\Delta \mathbf{h}_u = 0$. For injected neighbors $u \in N'_t(v) \setminus N_t(v)$, their features $\mathbf{h}_u$ are initialized to zero, i.e., $\mathbf{h}_u = 0$ and $\Delta \mathbf{h}_u = \mathbf{h}_u'$. Accordingly, Eq. (9) can be expressed as:

$$\Delta \mathbf{f}_t = \sum_{u \in N_t(v)} \Delta w_{u,t} \mathbf{h}_u^{k-1} + \sum_{u \in N'_t(v) \setminus N_t(v)} w_{u,t} \mathbf{h}_u'^{k-1}. \quad (10)$$

To illustrate, we begin with attacking a single-layer GNN (i.e., $k = 1$). In this case, our aim is to maximize $\Delta \mathbf{h}_v^1$ that corresponds to $\sum_{t=0}^1 \mathbf{p}_{t,1} \Delta \mathbf{f}_t$ according to Eq. (7). As $\Delta \mathbf{f}_0 = \Delta \mathbf{h}_u^0$ according to Eq. (10), we only need to maximize $\Delta \mathbf{f}_1$ as:

$$\Delta \mathbf{f}_1 = \sum_{u \in N_1(v)} \Delta w_{u,1} \mathbf{h}_u^0 + \sum_{u \in N'_1(v) \setminus N_1(v)} w_{u,1} \mathbf{h}_u'^0. \quad (11)$$

In Eq. (11), the first term corresponds to modifying the weights of original neighbors, whose features $\mathbf{h}_u^0$ remain fixed, and whose perturbation space of weights $\Delta w_{u,1}$ is tightly constrained by the graph structure, and the second term corresponds to the contribution of injected neighbors, whose weights $w_{u,1}$ and features $\mathbf{h}_u'^0$ can be freely manipulated.

During endpoint determination stage, although the features of injected flows are randomly initialized, an adversary can still maximize the adversarial impact by establish connections with original nodes to amplify the weight factor $\sum w_{u,1}$, and to perturb $\Delta \mathbf{f}_1$ as much as possible. Following the common choice in GraphSAGE [9], the weight $w_{u,1}$ is approximated by the mean aggregator as:

$$w_{u,1} = \frac{1}{d_v}, u \in N_1(v), \quad (12)$$

where d_v denotes the degree of node v .

To quantify the susceptibility of each node v in the line graph, we define a susceptible score λ_v by combining three key factors: prediction confidence, prediction correctness, and approximated weights. Formally, λ_v is computed as:

$$\lambda_v = (1 - p_v) \times \mathbb{1}(\arg \max(p_v) = y_v) \times \frac{1}{d_v}, \quad (13)$$

where p_v denotes the prediction confidence for node v , y_v denotes the ground-truth label for node v , the term $1 - p_v$ prioritizes nodes with lower prediction confidence, the indicator function $\mathbb{1}(\cdot)$ masks misclassified nodes, ensuring that only correctly classified nodes are considered for potential perturbation, and the term $\frac{1}{d_v}$ favors less-connected nodes.

A higher λ_v indicates that node v is correctly classified but susceptible due to low prediction confidence and weak structural robustness. By ranking candidate flows according to λ_v , we identify susceptible flows as anchors to determine feasible endpoints for injected adversarial flows. The assumptions about the adversary's knowledge are discussed in Appendix B.

3.4 Feature Optimization

Once the endpoints of injected flows have been determined, we derive the modified adjacency matrix A' . The next step is to optimize their features X' to maximize the adversarial impacts while maintaining stealthiness.

The features of injected flows X' can be updated by gradient descent using Eq. (3) as:

$$X'^* = \arg \min_{X'} \mathcal{L}_{\text{atk}}(f_{\theta^*}(A', X')), \text{ s.t. } \|X' - X\| \leq \epsilon. \quad (14)$$

However, the optimization process tend to reduce the similarity between the injected and the target flows, severely breaking the node homophily (i.e., similar nodes are more likely to connect). Following the typical definition, the homophily of node v can be measured as the similarity between its features X_v and the aggregated features of its neighbors as:

$$h_v = \text{sim}(c_v, X_v), c_v = \sum_{u \in N(v)} \frac{1}{\sqrt{d_v} \sqrt{d_u}} X_u, \quad (15)$$

where $\text{sim}(\cdot)$ denotes the cosine similarity metric.

If the injected flows excessively deviate from their related flows, they can be easily detected by simple homophily checks. Obviously, Eq. (3) can not guarantee homophily, since it optimize towards maximizing the adversarial impact by minimizing the homophily of the target nodes. To address this, we update Eq. (3) to integrate the homophily constraint for GIANT as:

$$X'^* = \arg \min_{X'} \mathcal{L}_{\text{atk}}(f_{\theta^*}(\mathcal{G}')) - \lambda C(\mathcal{G}, \mathcal{G}'), \quad (16)$$

where $\lambda \geq 0$ controls the trade-off, and $C(\mathcal{G}, \mathcal{G}')$ denotes a homophily regularization term that can be expressed as:

$$C(\mathcal{G}, \mathcal{G}') = \frac{1}{|V_{\text{atk}}|} \sum_{v \in V_{\text{atk}}} h_v \quad (17)$$

By incorporating this constraint, GIANT generates adversarial flows that not only maximize the adversarial impact, but also preserve structural consistency, thereby maintaining stealthiness.

Subsequently, the injected adversarial flows can be transferred to deceive the target GNIDS. The detailed attack process of GIANT is presented in Appendix C.

4 Evaluation

4.1 Experimental Setup

Datasets. We evaluate the attack performance of GIANT against several state-of-the-art GNIDS methods on two widely-used public datasets: 1) ToN-IoT dataset [19] comprises labeled network traffic collected from IoT/IIoT devices, capturing both benign activities and diverse attack scenarios, which has been widely adopted to evaluate AI-based security systems in IoT and edge environments; and 2) CIC-IDS2018-v2 [25] dataset was generated on the AWS cloud platform using a systematic, profile-driven approach to simulate realistic modern network traffic, including seven major attack types, which serves as a comprehensive benchmark dataset for network intrusion detection systems. For each dataset, we consider an IP address as compromised if it appears as a source IP address in malicious network flows. The attacker is only allowed to initiate adversarial network flows from compromised hosts to satisfy the strict constraint for practical attacks.

We use the `train_test_network` part of the ToN-IoT dataset, where each subset is formed by combining 20,000 malicious flows from a specific attack scenario and 300,000 benign flows, yielding a 1:15 malicious-to-benign ratio, maintaining a data imbalance in line with real-world environments. For the CIC-IDS2018-v2 dataset, we adopt the same sampling strategy to form each subset for each attack type. For both datasets, 80% of the flows are used for training, and the remaining 20% are reserved for testing and evaluation. The statistics of the datasets are displayed in Table 4-5 in Appendix D.

Constraints. We emphasize three implementation details relevant to constraints. First, the budget is specified exclusively over injected nodes/flows rather than edges as they are determined implicitly by the endpoints of flows in line graphs. Second, endpoint determination requires degree analysis and therefore assumes that the corresponding endpoints are observable. Specifically, an *on-path* adversary has full knowledge of the graph topology and may select anchors anywhere in the graph. In contrast, an *off-path* adversary only observes the partial topology involving compromised hosts and is thus limited to selecting anchors among flows where the compromised hosts participate. Finally, the features of injected flow are constrained within a plausible range to preserve structural consistency with related traffic, which prevents injected flows from being trivially removed by simple anomaly or similarity filters.

Metrics. We first evaluate the detection performance of various GNIDS models in the absence of perturbations using Macro-F1 score (F1) and the Area Under Curve (AUC). We then assess attack effectiveness using the accuracy reduction. As GIANT does not seek to alter existing traffic data, but to inject adversarial flows to mislead the decisions on the original benign and malicious flows.

GIANT Settings. We compare GIANT against four baseline attack methods, including PGD [18], Random Benign (R&B) [35], Random Malicious (R&M) [35], and TDGIA [43]. We use a two-layer GraphSAGE with LayerNorm on line graph as the surrogate model and transfer the injected adversarial flows to four baseline defense methods covering three graph construction modes including E-GraphSAGE [17], TS-IDS [21], Anomal-E [2] and E-ResGAT [3], integrated with a homophily filter [40]. A detailed description is provided in Appendix E.

4.2 Overall Results

4.2.1 Detection Performance Evaluation. Table 6 in Appendix F shows the detection performances of all baseline defense methods on two datasets. Overall, all evaluated GNIDS demonstrate strong detection performances across diverse attack scenarios in both IoT and cloud environments, indicating their general effectiveness in network intrusion detection. Among them, TS-IDS and Line-GraphSAGE achieve the most consistent results, while Anomal-E performs relatively weaker than others, likely due to its host-level graph construction mode, which provides coarser granularity and makes it harder to distinguish flows between different endpoints. In this work, we focus on evaluating the robustness of GNIDS rather than improving their detection performance.

4.2.2 Attack Effectiveness Evaluation. We perform a series of attack methods against the surrogate model (i.e., LineGraphSAGE) and then transfer the injected adversarial flows to multiple defense methods for effectiveness and transferability analysis. Table 1-2 summarize the effectiveness of all baseline attack methods against various target GNIDS on two datasets when injecting 10,000 adversarial flows.

Overall, GIANT, TDGIA, and R&M exhibit the most significant accuracy reduction, demonstrating their stronger capability to deceive GNIDS models. Although R&M follows random injection strategies, it achieves much higher attack effectiveness than PGD and R&M due to the inherent dataset imbalance, where injected flows are more likely to establish connections with benign flows and introduce larger topological perturbations.

Additionally, the effectiveness of GIANT varies across attack scenarios, requiring the adversary to adjust its perturbation strategy according to the underlying topological features. Interestingly, GIANT (on-path) only slightly outperforms GIANT (off-path), suggesting that even partial topological knowledge is sufficient to launch comparably effective attacks. This finding highlights the practicality of GIANT in real-world settings, where adversaries may only have local visibility of the network.

It is also notable that attacks tend to be more effective in IoT environments than in cloud environments, which can be attributed to the inherent sparsity and lower connectivity of IoT network topologies. Among the evaluated defense methods, TS-IDS and Anomal-E demonstrate relatively stronger robustness but still suffers significant performance degradation, indicating that current GNIDS solutions remain vulnerable to adaptive adversarial attacks.

4.2.3 Attack Efficiency Evaluation. To evaluate the efficiency of baseline attack methods, we inject adversarial flows in varying quantities [1000,2000,5000,10000] against the surrogate model. As illustrated in Figure 3, GIANT, TDGIA, and R&M outperform other baselines across most attack scenarios, with GIANT exhibiting the highest effectiveness, particularly on the ToN-IoT dataset. The performance gap between on-path and off-path settings is minor, suggesting consistent attack effectiveness under both visibility conditions.

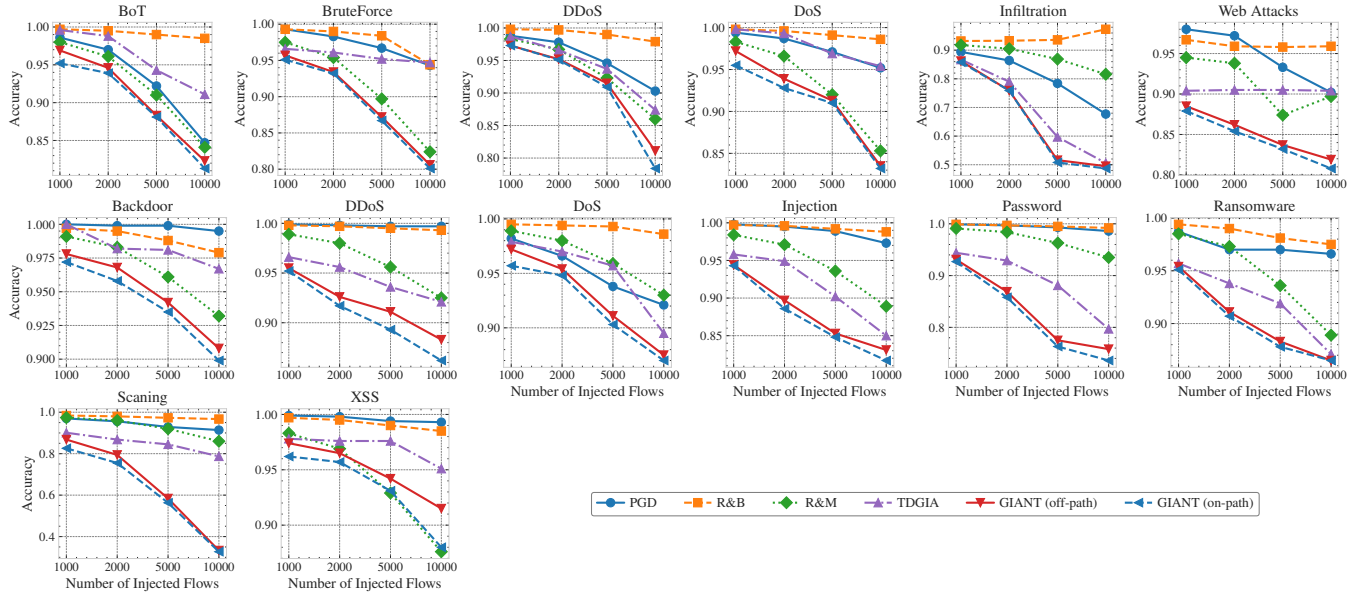
Notably, GIANT maintains a strong attack capability even when only a small number of adversarial flows are injected, demonstrating its high efficiency and adaptability. As the number of injected flows increases, the accuracy degradation becomes more evident,

Table 1: Attack effectiveness evaluation of baseline attack methods on the CIC-IDS2018-v2 dataset.

Attack	Defense	BruteForce	Bot	DDoS	DoS	Infiltration	Web Attacks
PGD	E-GraphSAGE	0.953 ▼ 1%	0.873 ▼ 13%	0.934 ▼ 7%	0.982 ▼ 2%	0.701 ▼ 23%	0.934 ▼ 6%
	TS-IDS	0.978 ▼ 2%	0.878 ▼ 12%	0.938 ▼ 6%	0.984 ▼ 2%	0.703 ▼ 28%	0.936 ▼ 6%
	Anomal-E	0.921 ▼ 2%	0.872 ▼ 6%	0.929 ▼ 6%	0.862 ▼ 1%	0.694 ▼ 17%	0.924 ▼ 4%
	E-ResGAT	0.966 ▼ 3%	0.866 ▼ 13%	0.922 ▼ 7%	0.967 ▼ 3%	0.693 ▼ 24%	0.919 ▼ 8%
R&B	E-GraphSAGE	0.962	0.979 ▼ 2%	0.988 ▼ 1%	0.999	0.919 ▼ 1%	0.991
	TS-IDS	0.977 ▼ 2%	0.999	0.989 ▼ 1%	0.999	0.982	0.992 ▼ 1%
	Anomal-E	0.912 ▼ 3%	0.921 ▼ 1%	0.972 ▼ 1%	0.846 ▼ 3%	0.855 ▼ 1%	0.963
	E-ResGAT	0.960 ▼ 4%	0.959 ▼ 4%	0.990 ▼ 1%	0.990 ▼ 1%	0.919 ▼ 2%	0.979 ▼ 2%
R&M	E-GraphSAGE	0.854 ▼ 11%	0.876 ▼ 12%	0.879 ▼ 12%	0.881 ▼ 12%	0.847 ▼ 8%	0.921 ▼ 7%
	TS-IDS	0.859 ▼ 14%	0.876 ▼ 12%	0.882 ▼ 12%	0.886 ▼ 11%	0.850 ▼ 13%	0.923 ▼ 8%
	Anomal-E	0.852 ▼ 9%	0.873 ▼ 6%	0.877 ▼ 11%	0.871	0.840 ▼ 2%	0.917 ▼ 5%
	E-ResGAT	0.839 ▼ 16%	0.857 ▼ 14%	0.876 ▼ 12%	0.869 ▼ 13%	0.836 ▼ 10%	0.912 ▼ 9%
TDGIA	E-GraphSAGE	0.946 ▼ 2%	0.935 ▼ 6%	0.897 ▼ 10%	0.982 ▼ 2%	0.526 ▼ 40%	0.926 ▼ 7%
	TS-IDS	0.979 ▼ 2%	0.940 ▼ 6%	0.906 ▼ 9%	0.984 ▼ 2%	0.527 ▼ 46%	0.939 ▼ 6%
	Anomal-E	0.936 ▼ 1%	0.932	0.896 ▼ 9%	0.846 ▼ 3%	0.524 ▼ 34%	0.926 ▼ 4%
	E-ResGAT	0.966 ▼ 3%	0.928 ▼ 7%	0.895 ▼ 10%	0.976 ▼ 2%	0.523 ▼ 41%	0.922 ▼ 8%
GIANT (off-path)	E-GraphSAGE	0.830 ▼ 13%	0.848 ▼ 15%	0.837 ▼ 16%	0.865 ▼ 14%	0.528 ▼ 40%	0.849 ▼ 15%
	TS-IDS	0.832 ▼ 17%	0.857 ▼ 14%	0.843 ▼ 16%	0.869 ▼ 13%	0.530 ▼ 46%	0.854 ▼ 15%
	Anomal-E	0.827 ▼ 12%	0.841 ▼ 9%	0.832 ▼ 15%	0.860 ▼ 1%	0.527 ▼ 34%	0.845 ▼ 12%
	E-ResGAT	0.827 ▼ 17%	0.839 ▼ 16%	0.829 ▼ 17%	0.853 ▼ 15%	0.523 ▼ 41%	0.842 ▼ 16%
GIANT (on-path)	E-GraphSAGE	0.824 ▼ 14%	0.839 ▼ 16%	0.815 ▼ 18%	0.863 ▼ 14%	0.519 ▼ 41%	0.833 ▼ 16%
	TS-IDS	0.829 ▼ 17%	0.842 ▼ 16%	0.820 ▼ 18%	0.863 ▼ 14%	0.521 ▼ 46%	0.843 ▼ 16%
	Anomal-E	0.822 ▼ 12%	0.835 ▼ 10%	0.803 ▼ 18%	0.859 ▼ 1%	0.513 ▼ 35%	0.829 ▼ 14%
	E-ResGAT	0.819 ▼ 18%	0.833 ▼ 16%	0.801 ▼ 19%	0.857 ▼ 14%	0.506 ▼ 43%	0.828 ▼ 17%

Table 2: Attack effectiveness evaluation of baseline attack methods on the ToN-IoT dataset.

Attack	Defense	Backdoor	DDoS	DoS	Injection	Password	Ransomware	Scanning	XSS
PGD	E-GraphSAGE	0.999 ▼ 7%	0.999 ▼ 7%	0.974 ▼ 2%	0.985 ▼ 1%	0.997	0.979 ▼ 2%	0.925 ▼ 6%	0.998
	TS-IDS	0.999	0.999	0.968 ▼ 3%	0.986 ▼ 1%	0.997	0.979 ▼ 2%	0.926 ▼ 7%	1.000
	Anomal-E	0.975	0.963 ▼ 1%	0.932 ▼ 5%	0.965 ▼ 1%	0.945 ▼ 1%	0.976	0.924 ▼ 5%	0.951 ▼ 2%
	E-ResGAT	0.999	0.997	0.930 ▼ 7%	0.981 ▼ 2%	0.994	0.975 ▼ 2%	0.922 ▼ 8%	0.980 ▼ 2%
R&B	E-GraphSAGE	0.988 ▼ 1%	1.000	0.980 ▼ 2%	0.978 ▼ 2%	0.992 ▼ 1%	0.988 ▼ 1%	0.977 ▼ 1%	0.996
	TS-IDS	0.992 ▼ 1%	0.999	1.000	0.999	0.999	0.988 ▼ 1%	0.980 ▼ 2%	0.997
	Anomal-E	0.968 ▼ 1%	0.959 ▼ 1%	0.959 ▼ 2%	0.957 ▼ 1%	0.929 ▼ 2%	0.956 ▼ 1%	0.970	0.954 ▼ 2%
	E-ResGAT	0.987 ▼ 1%	0.999	0.998	0.996	0.980 ▼ 2%	0.984 ▼ 1%	0.975 ▼ 2%	0.994
R&M	E-GraphSAGE	0.944 ▼ 6%	0.939 ▼ 6%	0.939 ▼ 6%	0.901 ▼ 10%	0.948 ▼ 5%	0.900 ▼ 10%	0.871 ▼ 12%	0.888 ▼ 11%
	TS-IDS	0.946 ▼ 5%	0.939 ▼ 6%	0.940 ▼ 6%	0.902 ▼ 10%	0.948 ▼ 5%	0.900 ▼ 10%	0.874 ▼ 13%	0.890 ▼ 11%
	Anomal-E	0.944 ▼ 4%	0.938 ▼ 3%	0.939 ▼ 4%	0.899 ▼ 7%	0.945	0.898 ▼ 8%	0.869 ▼ 10%	0.886 ▼ 9%
	E-ResGAT	0.940 ▼ 6%	0.933 ▼ 6%	0.938 ▼ 6%	0.897 ▼ 10%	0.944 ▼ 5%	0.897 ▼ 10%	0.869 ▼ 13%	0.885 ▼ 11%
TDGIA	E-GraphSAGE	0.976 ▼ 2%	0.932 ▼ 7%	0.905 ▼ 9%	0.862 ▼ 14%	0.806 ▼ 19%	0.881 ▼ 12%	0.799 ▼ 19%	0.964 ▼ 3%
	TS-IDS	0.980 ▼ 2%	0.933 ▼ 7%	0.908 ▼ 9%	0.862 ▼ 14%	0.807 ▼ 19%	0.885 ▼ 12%	0.800 ▼ 20%	0.965 ▼ 4%
	Anomal-E	0.976	0.931 ▼ 4%	0.905 ▼ 7%	0.860 ▼ 11%	0.806 ▼ 14%	0.881 ▼ 10%	0.799 ▼ 17%	0.961 ▼ 1%
	E-ResGAT	0.976 ▼ 2%	0.930 ▼ 7%	0.905 ▼ 9%	0.860 ▼ 14%	0.805 ▼ 19%	0.880 ▼ 12%	0.796 ▼ 20%	0.960 ▼ 1%
GIANT (off-path)	E-GraphSAGE	0.919 ▼ 8%	0.894 ▼ 10%	0.886 ▼ 11%	0.843 ▼ 15%	0.771 ▼ 23%	0.877 ▼ 12%	0.345 ▼ 64%	0.927 ▼ 7%
	TS-IDS	0.919 ▼ 8%	0.896 ▼ 10%	0.888 ▼ 11%	0.844 ▼ 16%	0.772 ▼ 23%	0.879 ▼ 12%	0.345 ▼ 66%	0.928 ▼ 7%
	Anomal-E	0.918 ▼ 6%	0.892 ▼ 8%	0.885 ▼ 9%	0.842 ▼ 13%	0.771 ▼ 18%	0.876 ▼ 10%	0.344 ▼ 63%	0.926 ▼ 5%
	E-ResGAT	0.918 ▼ 8%	0.891 ▼ 11%	0.884 ▼ 12%	0.840 ▼ 16%	0.770 ▼ 23%	0.875 ▼ 12%	0.343 ▼ 66%	0.925 ▼ 7%
GIANT (on-path)	E-GraphSAGE	0.909 ▼ 9%	0.873 ▼ 12%	0.883 ▼ 11%	0.830 ▼ 17%	0.749 ▼ 25%	0.876 ▼ 12%	0.337 ▼ 65%	0.891 ▼ 11%
	TS-IDS	0.911 ▼ 9%	0.874 ▼ 13%	0.884 ▼ 12%	0.830 ▼ 17%	0.749 ▼ 25%	0.879 ▼ 12%	0.340 ▼ 66%	0.893 ▼ 11%
	Anomal-E	0.909 ▼ 7%	0.872 ▼ 10%	0.879 ▼ 10%	0.828 ▼ 14%	0.747 ▼ 20%	0.875 ▼ 10%	0.336 ▼ 64%	0.889 ▼ 8%
	E-ResGAT	0.908 ▼ 9%	0.872 ▼ 13%	0.878 ▼ 12%	0.828 ▼ 17%	0.745 ▼ 25%	0.875 ▼ 12%	0.336 ▼ 66%	0.889 ▼ 11%

**Figure 3: Accuracy reduction of various attacks against the surrogate model (i.e., LineGraphSAGE).**

further amplifying the advantage of GIANT over random or heuristic strategies. In contrast, the performance of random strategies (e.g., PGD, R&B, R&M) fluctuates considerably with varying injection scales, indicating their instability and lack of robustness.

It is worth noting that in certain scenarios (e.g., Scanning in the ToN-IoT dataset), GIANT achieves superior attack effectiveness, likely due to the sparse connectivity diversity in the graph.

4.3 Ablation Studies

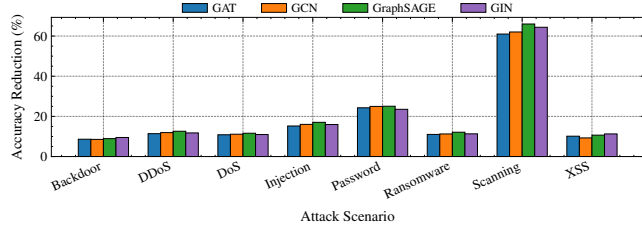
we conduct ablation studies on the ToN-IoT dataset to evaluate the contributions of key components in GIANT. We use "w/o AS" and "w/o HC" to refer to the variants of GIANT that removes the attack strategy during the endpoint determination phase and the homophily constraint during the feature optimization phase. All

variants are evaluated under both off-path and on-path assumptions. Table 3 summarizes the comparison results. Overall, the on-path setting of GIANT slightly outperforms the off-path one, and both the attack strategy and homophily constraint prove to be essential components. Removing the homophily constraint leads to the most significant accuracy reduction, as all GNIDS solutions are integrated with a homophily filter, therefore adversarial flows with anomalous features are directly filtered out. Also, removing the attack strategy weakens GIANT's ability to select susceptible flows as targets, reducing the overall attack effectiveness. The above ablation studies confirm the contributions of the two key components.

Additionally, we investigate the influence of different surrogate models. Figure 4 illustrates the transferability of GIANT against

Table 3: Ablation studies of GIANT on the ToN-IoT dataset.

Assumption	Variant	Backdoor	DDoS	DoS	Injection	Password	Ransomware	Scanning	XSS
Off-Path	w/o AS	0.964 ▼ 4%	0.947 ▼ 5%	0.913 ▼ 8%	0.872 ▼ 13%	0.795 ▼ 21%	0.891 ▼ 11%	0.547 ▼ 44%	0.974 ▼ 3%
	w/o HC	0.976 ▼ 2%	0.938 ▼ 6%	0.915 ▼ 8%	0.892 ▼ 11%	0.814 ▼ 19%	0.925 ▼ 7%	0.702 ▼ 29%	0.946 ▼ 5%
	GIANT	0.908 ▼ 9%	0.883 ▼ 12%	0.875 ▼ 12%	0.831 ▼ 17%	0.758 ▼ 24%	0.865 ▼ 13%	0.333 ▼ 66%	0.915 ▼ 8%
On-Path	w/o AS	0.960 ▼ 4%	0.935 ▼ 6%	0.889 ▼ 11%	0.855 ▼ 14%	0.788 ▼ 21%	0.886 ▼ 11%	0.513 ▼ 48%	0.952 ▼ 5%
	w/o HC	0.973 ▼ 3%	0.934 ▼ 6%	0.907 ▼ 9%	0.864 ▼ 14%	0.793 ▼ 21%	0.873 ▼ 13%	0.699 ▼ 29%	0.937 ▼ 6%
	GIANT	0.899 ▼ 10%	0.862 ▼ 14%	0.870 ▼ 13%	0.817 ▼ 18%	0.736 ▼ 26%	0.865 ▼ 13%	0.327 ▼ 66%	0.880 ▼ 12%

**Figure 4: Accuracy Reduction of GIANT on the ToN-IoT dataset using different surrogate models against TS-IDS.**

TS-IDS, the most robust defense methods. GraphSAGE achieves the most accuracy reduction. Thus we adopt it as the surrogate model.

5 Related work

Graph-based Network Intrusion Detection. Recent advances in Graph Neural Networks (GNNs) [22] have inspired their adoption in network intrusion detection systems [36]. E-GraphSAGE [17] and TS-IDS [21] leverage inductive GNNs to model endpoint-level interactions, enhancing generalization to unseen nodes. TCG-IDS [38], 3D-IDS [23] and Netvigil [13] extends this idea by integrating temporal and structural dependencies in dynamic traffic graphs. Anomal-E [2] instead focuses on host-level interactions to capture coarse-grained communication behaviors, while ArganIDS [33] and E-ResGAT [3] employ line graph representations to directly characterize flow-level interactions. Though promising, the robustness of GNIDS remains largely unexplored.

Graph Adversarial Attacks. Graph adversarial attacks can be broadly categorized into graph modification and graph injection attacks [29]. Modification-based methods, such as Nettack [44] and Metattack [45], directly perturb the graph structure or node features to mislead GNN predictions. In contrast, injection-based methods like TDGIA [43] and HAO-GIA [4] introduce new nodes or edges to craft adversarial subgraphs, offering higher practicality in real-world systems where direct graph modification is infeasible. Our work builds on this line of research by proposing a general and practical injection attack framework tailored to GNIDS, emphasizing both attack realism and transferability.

6 Discussion

In this section, we elaborate on the attack strategy, efficiency considerations and potential extensions, as well as future work.

Attack strategy. A critical factor of GIANT lies in selecting vulnerable flows as targets during the endpoint determination phase. The connection establishment process can manage a trade-off between depth and width by repeatedly targeting the same flows or spreading across diverse flows. Specifically, the adversary may maximize effectiveness by focusing merely on the most vulnerable flows

or broaden coverage by distributing across multiple moderately vulnerable ones. In this work, we focus more on broadening coverage. Future work could explore a more effective attack strategy to strike a balance between effectiveness and efficiency.

Efficiency considerations. Though the low computational complexity of GIANT allows it to scale to large graphs, the line graph construction remains resource-intensive. To handle the large graph incurred by the massive traffic, flow aggregation and host-level interaction summaries can be leveraged. Instead of operating on individual flows, future work could explore adversarial injections targeting aggregated flows or summarized host interactions, reducing computational overhead while maintaining attack effectiveness.

Extension to GIDS. In this work, we evaluate GIANT in the context of GNIDS. However, the design of GIANT is general enough to extend to GHIDS for provenance graph analysis [37] and lateral movement detection [16]. In these applications, adversarial flow injections can be naturally mapped to the insertion of synthetic audit/network logs [8] that mimic benign entity dependencies or host interactions. We consider further exploring GIANT's potential impact across diverse IDS paradigms as future work.

7 Conclusion

This work proposes GIANT, a practical and general adversarial attack framework for comprehensive robustness evaluation of GNIDS. Through extensive experiments on two public datasets spanning IoT and cloud environments, we demonstrate the superiority of GIANT compared with existing attacks methods in effectiveness, efficiency and transferability. Through GIANT, we validate critical vulnerabilities in state-of-the-art GNIDS, while also revealing the deficiencies of prior attack methods in simulating practical evasion techniques. In summary, GIANT shows promising potential for GNIDS evaluation, alleviating the concerns of security professionals on the reliability of GNIDS in real-world deployments. Furthermore, we emphasize the necessity for continued efforts in strengthening GNIDS against adversarial attacks, enabling more reliable network intrusion detection in security-critical infrastructures.

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A Design Goals

Effectiveness. GIANT is designed to effectively compromise GNIDS by injecting adversarial flows to mislead the target model into misclassification. The effectiveness is usually measured by the misclassification rate of the original flows under the influence of the injected flows.

Efficiency. GIANT aims to inject as few adversarial flows as possible while perturbing as many original flows as possible, leveraging the aggregation dynamics of GNNs to maximize the adversarial impact.

Practicality. GIANT is developed under realistic assumptions without access to model parameters or graph construction mode. It merely injects new adversarial flows through compromised hosts rather than modify existing traffic data. Moreover, it works in both on-path cases with global visibility and off-path with local visibility, aligning with real-world constraints.

Stealthiness. GIANT requires injected flows to remain stealthy, ensuring that they do not significantly differ from associated flows, so as to avoid being easily filtered by similarity analysis.

Scalability. GIANT follows a two-phase process: heuristic degree analysis for endpoint determination, followed by gradient-guided and homophily-aware feature optimization. This decomposition keeps the computational complexity far lower than model training, ensuring scalability to large-scale graphs and various target models.

Transferability. GNIDS adopt diverse graph construction modes, posing challenges for attack transferability. GIANT should maintain effective across different graph construction modes, making it broadly applicable to various GNIDS.

B Assumptions

As the adversary's knowledge directly shapes the strategy for selecting susceptible flows, we consider two typical assumptions widely adopted in network security analysis:

On-Path Assumption (global visibility). The adversary resides on the communication path, observing all network traffic data and thus knowing the complete graph topology. Consequently, λ_v can be computed globally, and susceptible flows can be freely selected as anchors.

Off-Path Assumption (local visibility). The adversary is located off the communication path, observing only the traffic data involving compromised hosts and therefore knowing only a partial graph topology, essentially the ego-network around compromised hosts. As a result, λ_v can be evaluated only for observed nodes/flows, and anchor selection is restricted to flows in which compromised hosts participate.

In summary, under the on-path assumption, anchors are chosen from the entire candidate set, whereas under the off-path assumption, anchors are restricted to the subset involving compromised hosts. The two cases are analyzed separately. It is worth noting that the endpoint determination strategy naturally generalizes to multi-layer GNNs [43].

C Attack Process

In GIANT, we employ a sequential attack strategy to inject adversarial flows in batches. In each batch, we add a limited number of

flows to the existing traffic data, determine their endpoints, and optimize their features. We repeat this process until the overall injection budget is exhausted. This sequential attack strategy allows the attack to exploit intermediate topological changes and often improves transferability. The overall attack process of GIANT is illustrated in Algorithm 1.

Algorithm 1: The attack process of GIANT

Input: original graph $\mathcal{G} = \{V, A, X\}$; surrogate model $\mathcal{M}$; batch B ; total budget b ; per-batch budget b_{seq} ; compromised hosts C .

Output: Modified graph $\mathcal{G}' = (V', A', X')$.

// Initialization

```

1 Generate  $\{\hat{y}_v\}_{v \in V}$  as ground-truth labels using  $\mathcal{M}(\mathcal{G})$ ;
2  $\mathcal{G}' \leftarrow \mathcal{G}$ ;
3 while  $b > 0$  do
4    $batch \leftarrow \min(b_{\text{seq}}, b)$ ;
5   // Endpoint Determination
6   for  $v \in V'$  do
7     Compute prediction confidence  $p_v \leftarrow \mathcal{M}(\mathcal{G}')$ ;
7     Compute susceptible score  $\lambda_v$  using Eq. (13);
8   Sort candidate flows by  $\lambda_v$  in descending order;
9   Select  $batch$  source endpoints from the sources of flows
    that belong to compromised hosts  $C$ ;
10  Select  $batch$  destination endpoints from the destinations
    of flows excluding selected endpoints;
11  Pair endpoints to form  $batch$  injected flows  $V_I$ ;
12  // Feature Optimization
13  Optimize injected flow features  $X_I$  using Eq. 16;
14  // Update Graph and Budget
15  Update  $V', A', X'$  with  $V_I, X_I$  using Eq. 2;
16   $\mathcal{G}' \leftarrow (V', A', X')$ ;
17   $b \leftarrow b - batch$ ;
18 return  $\mathcal{G}'$ ;
```

Complexity Analysis. Given a surrogate model $\mathcal{M}$ with computational complexity T . For typical GNNs, $T = O(ED)$, where E denotes the number of edges and D denotes the node feature dimension. The endpoint determination phase requires one inference of $\mathcal{M}$ to compute p_v , that costs $O(T)$, and the computation of λ_v that costs $O(E)$, which is asymptotically negligible. Let Δ_S be the number of optimization epochs and B be the number of batches in sequential injection, thus the complexity of the feature optimization phase is $O(\Delta_S BT)$. The overall complexity for GIANT is $O(\Delta_S BT)$.

In practice, $\Delta_S B$ is typically much smaller than the training epochs of $\mathcal{M}$, so the attack process incurs significantly less computational overhead than model training. Overall, the two-phase attack process of GIANT enables the scalability to both large-scale graphs and diverse target models.

D Dataset

We provide a detailed description of the two datasets in Table 4-5.

Table 4: Overview of the ToN-IoT dataset.

Class	# Flow	# Compromised IP	# Compromised Endpoint
Backdoor	20000	3	828
DDoS	20000	4	16229
DoS	20000	4	1666
Injection	20000	3	15172
Password	20000	2	15187
Ransomware	20000	5	1411
Scan	20000	4	1985
XSS	20000	2	9944
Benign	300000	-	-

Table 5: Overview of the CIC-IDS2018-v2 dataset.

Class	# Flow	# Compromised IP	# Compromised Endpoint
BruteForce	20000	10	13778
Bot	20000	2	19916
DoS	20000	10	12338
DDoS	20000	4	18800
Infiltration	20000	198	14363
WebAttacks	20000	6	439
Benign	300000	-	-

E GIANT Settings

Baseline Attack Methods. We compare our GIANT attack with different baseline attack methods, including PGD, Random Benign (R&B), Random Malicious (R&M) and TDGIA methods, where PGD, R&B and R&M are adapted to GIA settings inline with the definition of our threat model. Specifically, we randomly select endpoints to determine the source and destination of adversarial flows, where the source endpoint shall be within compromised hosts. For PGD, we optimize the flow features with projected gradient descent as a universal first-order adversary. For R&B and R&M, we randomly sample benign and malicious flows to directly utilize their features. Notably, TDGIA is also adapted to satisfy the inherent graph construction rules and practical constraints.

Surrogate Attack Model. GraphSAGE is the first inductive GNN model with great generalizability to unseen nodes and large graphs, which is widely adopted in existing GNIDS solutions (e.g., E-GraphSAGE and LineGraphSage). In this work, we represent network flow logs as line graph format, build a two-layer GraphSAGE (with LayerNorm) model as the surrogate attack model, perform GIANT and other attack methods on it, and transfer the newly-added adversarial network flows to all defense methods.

Baseline Defense Methods. We evaluate the performance of GIANT and other baseline attack methods on current GNIDS solutions with different graph construction modes, which can be divided into three categories: 1) endpoint graph (i.e., E-GraphSAGE, TS-IDS), where each node represents an IP-port pair connected by flows; 2) host graph (i.e., Anomal-E), where each node represents an IP addresses connected by flows; and 3) line graph (i.e., E-ResGAT, LineGraphSAGE), where each node represents a network flow, and edges are established following heuristic rules. Note that all baseline defense methods are integrated with a homophily defender to

directly remove those highly suspicious flows that greatly break the homophily.

F Detection Performance Evaluation

Table 6 summarizes the detection performances of all baseline defense methods on the ToN-IoT and CIC-IDS2018-v2 datasets. The results indicate that GNIDS achieve better detection performance in IoT environments compared with cloud environments, likely due to the sparser connectivity and more predictable communication patterns of IoT devices.

Table 6: Detection performance evaluation of baseline defense methods.

Defense Methods		E-GraphSAGE		TS-IDS		Anomal-E		E-ResGAT		LineGraphSAGE	
Dataset	Attack	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC
ToN-IoT	Backdoor	0.999	1.000	1.000	1.000	0.894	0.950	0.998	0.996	0.999	1.000
	DDoS	0.990	0.998	1.000	1.000	0.847	0.973	0.989	0.999	0.995	0.999
	DoS	0.988	0.999	0.999	1.000	0.893	0.925	1.000	0.999	0.992	1.000
	Injection	0.995	0.999	1.000	1.000	0.866	0.973	0.999	0.999	0.996	1.000
	Password	0.993	1.000	1.000	1.000	0.798	0.962	0.999	0.999	0.999	1.000
	Ransomware	0.998	1.000	1.000	1.000	0.893	0.989	0.999	0.999	0.997	1.000
	Scanning	0.956	0.999	1.000	1.000	0.899	0.990	1.000	0.999	0.967	0.998
	XSS	0.994	1.000	1.000	1.000	0.848	0.943	1.000	1.000	0.998	1.000
	Average	0.989	0.999	1.000	1.000	0.867	0.963	0.998	0.999	0.993	1.000
CIC-IDS2018-v2	BruteForce	0.906	0.998	1.000	1.000	0.829	0.979	0.992	0.990	0.999	0.999
	Bot	0.993	1.000	1.000	1.000	0.807	0.975	0.992	0.992	1.000	1.000
	DDoS	0.999	1.000	1.000	1.000	0.934	0.998	0.979	0.976	1.000	1.000
	DOS	0.999	1.000	1.000	1.000	0.474	0.424	0.992	0.992	1.000	1.000
	Infiltration	0.662	0.768	0.938	0.994	0.474	0.429	0.497	0.559	0.699	0.814
	Web Attacks	0.983	0.997	1.000	1.000	0.881	0.980	0.963	0.940	0.994	0.999
	Average	0.924	0.961	0.990	0.999	0.733	0.798	0.903	0.908	0.949	0.969

G Case Study

We notice that on the ToN-IoT dataset, various attacks perform significantly better in the Scanning scenario than in other cases. This can be attributed to the distinct interaction patterns of scanning attack traffic, where numerous low-degree malicious flows connect sparsely to a few high-degree benign hubs. Such asymmetric connectivity creates a hub-spoke structure that amplifies the impact of small perturbations, rendering GNIDS more vulnerable to topological perturbations. Table 7 reports the degree analysis of various attack scenarios of the ToN-IoT dataset.

Table 7: Degree analysis of various attack scenarios of the ToN-IoT dataset.

Scenario	Max Degree of Benign	Mean Degree of Benign	Max Degree of Malicious	Mean Degree of Malicious
Backdoor	9172	3393.35	2174	2050.74
DDoS	6565	2903.27	1960	1748.23
DoS	7867	3403.98	7849	69.16
Injection	7786	3407.33	1621	1054.18
Password	9250	3438.75	2219	1237.59
Ransomware	7861	3413.43	2863	2049.68
Scanning	7842	3371.32	320	85.40
XSS	7785	3391.82	1336	973.08

These results suggest that attacks exploiting degree asymmetry and cross-class edge manipulations can achieve higher effectiveness with minimal modifications, offering insights into the design of topology-aware attack strategies.